A Dynamic Model of Stellar Populations

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CHAPTER 1: INTRODUCTION

A. Research Question

Modeling stellar evolution is a significant aspect of astronomy and astrophysics (Gallart *et al.* 2005). Stellar models help us understand the dynamics of the universe, make predictions on star formation, test theoretical frameworks, and inform observational strategies (Aguirre 2018, Bowman *et al.* 2023). Numerous stellar models have been proposed (see for example Verro *et al.* (2022)), these models are primarily purely theoretical, or are based on individuals stars instead of entire populations.

The life cycle of a star is primarily determined by its mass. Stars are formed from the gravitational collapse of material in the interstellar medium (ISM). Once a star is born, the fusion of elements in its core releases energy and generates outward pressure which prevents further gravitational collapse, maintaining a state known as hydrostatic equilibrium (Lang 2013) (also seen in Eq. (3)). Eventually, the star runs out of elements in its core to fuse and will die, as hydrostatic equilibrium can no longer be maintained (Lang 2013). Although a high mass star has more fuel to burn, it has a shorter lifetime because the rate of fusion in its core is much higher (Lang 2013). Such a star will die in a highly energetic explosions known as a supernova. This causes the violent ejection of the star's outer layers back into the ISM, while the core collapses into a 'stellar remnant', in this case either a neutron star or a black hole (Lang 2013). A low mass star like the Sun, on the other hand, will have a much less violent death, shedding off its outer layers into the ISM, while its core collapses into a less extreme stellar remnant called a white dwarf. The component of the star that returns to the ISM can be reformed into stars later on in the universe, whereas the component that forms into a stellar remnant cannot reform into stars (Lang 2013).

While much research has been done into stellar evolution, limitations in technology and the large scope of the field has left many open questions (Steindl 2022). A significant gap in extant research is in developing mathematical models to describe the distribution of stellar masses over time, accounting for the entire process outlined above. Modeling the evolution of populations through time is a particularly well studied task in the field of epidemiology and ecology. A cornerstone method of mathematical modeling in epidemiology and ecol-

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ogy is the "compartment model framework" in which different populations are modeled as different compartments, and the "flow" of agents from each population is modeled via a rate equation (Rothman *et al.* 2012, Ledder 2023). **The main aim of this paper is to adapt compartment methods from epidemiology and ecology to model how the distribution of stellar masses evolves over time.** Doing so would give us insights into

- the rate, f, at which mass from the interstellar medium reforms into stars
- what the typical fraction p_{ret} of a star's mass that returns to the interstellar medium is, as opposed to forming stellar remnants
- how low and high mass stars compare, and how their distribution changes over time

We propose a novel system of differential equations to model the evolution of a stellar population based on the life cycles of stars as outlined above, using the initial conditions in which galaxies formed from the gravitational collapse of gas in the early Universe - about 250 million years (Myr) after the Big Bang (Hashimoto *et al.* 2018), which itself occurred 13.8 billion years (Gyr) ago (Planck Collaboration 2020). From this model we predict the observational properties of galaxies of different ages, determine f and p_{ret} by fitting the model to real observations of galaxies, and examine the stellar mass distribution.

B. Background Information

1. The Initial Mass Function

The empirically-determined distribution of masses among stars when formed is known as the initial mass function (IMF). It is often given as a probability density function, denoted $\Phi(m)$. The IMF was first described by Salpeter (1955), who determined $\Phi(m) \propto m^{-2.35}$ for $0.4\,M_\odot < m < 10\,M_\odot$, where $M_\odot$ is the mass of the Sun. This was modified by Miller and Scalo (1979), who discovered that it more closely matches a constant below $1\,M_\odot$. Kroupa (2001, 2002) later found evidence that the IMF shows remarkable uniformity between different types of stellar populations, including "present-day star formation in small molecular clouds, rich and dense massive star-clusters forming in giant clouds, through to ancient and

metal-poor exotic stellar populations". The Kroupa IMF takes the form

$$\Phi(m) = \begin{cases} c_0 m^{-\alpha_0}, & 0.01 < m/M_\odot \leq 0.08 \\ c_1 m^{-\alpha_1}, & 0.08 < m/M_\odot \leq 0.50 \\ c_2 m^{-\alpha_2}, & 0.50 < m/M_\odot \leq 1.00 \\ c_3 m^{-\alpha_3}, & 1.00 < m/M_\odot \end{cases} \tag{1}$$

where

$$\alpha_0 = 0.3 \pm 0.7$$
$$\alpha_1 = 1.3 \pm 0.5$$
$$\alpha_2 = 2.3 \pm 0.3$$
$$\alpha_3 = 2.3 \pm 0.7$$

with the numbers following $\pm$ indicating uncertainty in the value. We require Φ to be continuous, so the values of c_1, c_2, and c_3 are determined by c_0 according to

$$\begin{aligned} c_1 &= 0.08^{\alpha_1 - \alpha_0} \cdot c_0 \\ c_2 &= 0.50^{\alpha_2 - \alpha_1} \cdot c_1 \\ &= 0.50^{\alpha_2 - \alpha_1} \cdot 0.08^{\alpha_1 - \alpha_0} \cdot c_0 \\ c_3 &= 1.00^{\alpha_3 - \alpha_2} \cdot c_2 \\ &= 0.50^{\alpha_2 - \alpha_1} \cdot 0.08^{\alpha_1 - \alpha_0} \cdot c_0. \end{aligned}$$

Since Φ is a probability density function, c_0 must then take a value such that the normalisation condition

$$\int_{m_{\min}}^{m_{\max}} \Phi(m)\, dm = 1 \tag{2}$$

is satisfied, where $m_{\min}$ and $m_{\max}$ are set by physical bounds on the mass of a star. Typically, $m_{\min} = 0.01\, M_\odot$ is used in the literature (e.g. Kroupa 2001, 2002) as a compact object less massive than this would be better classified as a planet (Seeds and Backman 2016). The upper bound on the mass of a star is not well constrained, being possibly up to about $200\, M_\odot$, above which a star can not grow because its outer layers would be held too loosely by the star's gravity and would be blown off its surface by the radiation it emits (Carroll and Ostlie 2007). However, Kroupa (2001) uses $m_{\max} = 50\, M_\odot$, which we adopt.

2. *Stellar Evolution*

Stars are formed from the gravitational collapse of material in the interstellar medium. When a star is born, it is in a category known as the main sequence. The main sequence is the phase of the life cycle of a star where specifically hydrogen (the lightest element) is fused into helium, to maintain hydrostatic equilibrium. Once a star has gone through all of it's hydrogen, it is no longer classified as a main sequence star, typically moving on into the category of red giants. The main sequence is the most important part of the life cycle of a star, as a star is in the main sequence for most of it's life cycle (Seeds and Backman 2016).

The main sequence (MS) lifetime of a star is empirically known to be

$$\frac{\tau}{\tau_\odot} \simeq \left(\frac{m}{M_\odot}\right)^{-2.5} \tag{3}$$

where the predicted lifetime of the Sun is $\tau_\odot \approx 10$ Gyr (Seeds and Backman 2016).

The mass-luminosity relation for stars on the MS is

$$\frac{L}{L_\odot} \simeq \left(\frac{m}{M_\odot}\right)^{3.5} \tag{4}$$

where $L_\odot$ is the luminosity of the sun (Seeds and Backman 2016).

A similar relationship holds for the radius of a MS star:

$$\frac{R}{R_\odot} \simeq \left(\frac{m}{M_\odot}\right)^{0.8} \tag{5}$$

where $R_\odot$ is the radius of the sun (Seeds and Backman 2016).

These are rough empirical fits which hold only for MS stars, which account for 90% of stars, but tend to be dimmer than the other 10% Seeds and Backman (2016).

3. *The Simple Stellar Population Model*

The existing literature on the evolution of stellar populations is fairly limited, focusing primarily on how changes in the spectra (the intensity as a function of wavelength of light) emitted by individual stars over their lifetimes affect the spectrum of the whole population, rather than how the distribution of stellar masses evolves over multiple generations of stars. Often, such as in Verro *et al.* (2022), a simple stellar population (SSP) is assumed, in which

there is no star formation following the initial creation of the population. Under this model, the stars of highest mass die first and are not replenished, creating a mass cutoff $m_{\text{high}}(t)$ in the population which decreases over time, $\frac{dm_{\text{high}}}{dt} < 0$, with $m_{\text{high}}(0) \leq m_{\text{max}}$. The mass function at time t for a simple stellar population can simply be written

$$\phi_{\text{SSP}}(m,t) = \frac{1}{c(t)} \begin{cases} \Phi(m), & m_{\text{min}} < m < m_{\text{high}}(t) \\ 0, & m > m_{\text{high}}(t) \end{cases} \tag{6}$$

where the function $c(t) = \int_{m_{\text{min}}}^{m_{\text{high}}(t)} \Phi(m')\, dm'$ in the denominator ensures the probability density function is normalised.

The spectrum of the population is then modelled by

$$\begin{aligned} f_{\text{SSP}}(t,Z) &= \int_{m_{\text{min}}}^{m_{\text{max}}} f_*(m,t,Z)\, \phi_{\text{SSP}}(m,t)\, dm \\ &= \int_{m_{\text{min}}}^{m_{\text{high}}(t)} f_*(m,t,Z)\, \Phi(m)\, dm, \end{aligned} \tag{7}$$

where

- Z is the metallicity, often measured by the concentration of iron relative to hydrogen, $Z = [\text{Fe/H}]$.
- $f_{\text{SSP}}(t,Z)$ is the spectrum of the whole population at time t since its formation, given an overall metallicity Z.
- $f_*(m,t,Z)$ is the theoretical spectrum of a star of mass m at age t with metallicity Z. In addition to the assumption that each star is the same age as the population, the SSP model also assumes that the metallicity of each star is the same as the overall metallicity of the population.

The assumptions of the SSP model severely limit its applicability, making it accurate only for populations with negligible rates of star formation such as elliptical galaxies (Kennicutt 1998), which make up a minority of the galaxies in the universe (Loveday 1996). For galaxies undergoing active star formation, such as gas-rich spiral galaxies (Kennicutt 1998), the SSP model fails to incorporate the presence of high-mass stars whose expected lifetimes are less than the age of the population. For example, when applied to the Milky Way (MW) galaxy, it would predict that the Sun should not exist as its predicted lifetime of 10 Gyr (Seeds and

Backman 2016) means that if it had formed at the same time as the MW, within the first billion years of the universe (Carroll and Ostlie 2007), it would have died about 3 billion years ago. The solution to this apparent problem is that the Sun was in fact formed well after the MW itself, being about 4 Gyr in age (Seeds and Backman 2016).

Verro *et al.* (2022) attempt to model the spectra of galaxies according to Eq. (7), focusing on an accurate determination of f_*, the spectra emitted by individual stars. They note that their model does not correctly predict some colours of galaxies[1], which, as in Eq. (7), also depend on the mass function. It is therefore plausible that the assumption of a simple stellar population was a major contributor to the errors in their predictions.

These issues demonstrate the need for a dynamic model of the distribution of stellar masses which incorporates ongoing star formation.

CHAPTER 2: MODEL

To improve upon the issues of the SSP model outlined in the previous chapter, we here present a compartment model which describes the evolution of a stellar population, incorporating star formation as well as star deaths. It describes the distribution of stellar masses, the total mass of the ISM associated with the population and the total mass of all stellar remnants formed from the population. Our model is given by

$$\frac{d\rho(m,t)}{dt} = f M_{\mathrm{ISM}}(t) \cdot \frac{m\Phi(m)}{\langle m \rangle_\Phi} - r(m)\,\rho(m,t) \tag{8}$$

$$\frac{dM_{\mathrm{ISM}}(t)}{dt} = -\,f M_{\mathrm{ISM}}(t) + p_{\mathrm{ret}} \int_{m_{\min}}^{m_{\max}} r(m)\rho(m,t)\,dm \tag{9}$$

$$\frac{dM_{\mathrm{R}}(t)}{dt} = (1-p_{\mathrm{ret}}) \int_{m_{\min}}^{m_{\max}} r(m)\rho(m,t)\,dm \tag{10}$$

where

- $\rho(m,t)$ is the total mass per unit mass of star, that is the combined mass of stars with individual masses in the range $[m, m+dm]$, divided by dm.
- $M_{\mathrm{ISM}}(t)$ is the total mass of the ISM.

[1] There are various ways of quantifying the colour of an object in astronomy, each of which can be derived from its spectrum; the model by Verro *et al.* (2022) correctly predicts some, but not all, of these colours.

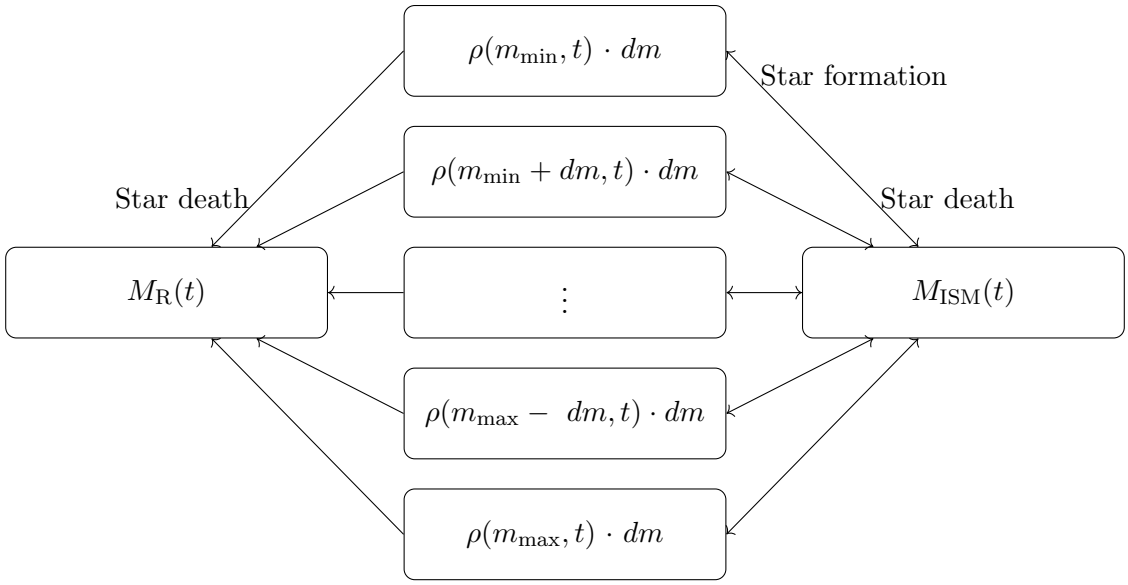


FIG. 1. In-depth diagram visualising binning of masses to encode continuous distribution into IDEs.

- $M_{\mathrm{R}}(t)$ is the total mass of all stellar remnants in the population, including white dwarfs, neutron stars, and black holes formed from the deaths of stars.

- $f \geq 0$ Gyr^{-1} is the fraction of the mass contained in the ISM that goes into star formation per unit time.

- $p_{\mathrm{ret}} \in [0, 1]$ is the proportion of a star's mass that is returned to the interstellar medium, rather than becoming a stellar remnant, upon the star's death.

- $r(m) = 1/\tau(m)$ is the rate at which stars of mass m die, having typical lifetimes $\tau(m) \simeq (m/M_{\odot})^{-2.5}\, \tau_{\odot}$ where the predicted lifetime of the Sun is $\tau_{\odot} \approx 10$ Gyr (Seeds and Backman 2016). We simplify the model by taking r to be independent of time.

- $\Phi(m)$ is the Kroupa (2001, 2002) initial mass function given in Eq. (1).

- $m_{\min}$ and $m_{\max}$ are the minimum and maximum masses, respectively, of stars in our model.

- $\langle m \rangle_{\Phi} = \int_{m_{\min}}^{m_{\max}} m' \Phi(m')\, dm'$ is the expected value of m under the IMF $\Phi(m)$, and serves as a normalisation factor.

Our model is based on the following theory and assumptions.

Stars die at some rate $r(m) = 1/\tau(m)$. As such, for a given mass of the star m, the total mass of stars that die per unit time per unit mass of the star is $r(m)\rho(m,t)$. The total mass of stars that die per unit time is therefore $\int_{m_{\min}}^{m_{\max}} r(m)\rho(m,t)dm$. We assume that some fixed proportion p_{ret} of a star's mass is returned to the ISM upon its death, and the remaining proportion becomes stellar remnants.

The amount of mass that forms into stars from the ISM per unit time is fM_{ISM}. We assume that the proportion of the ISM that forms into stars of a specific mass is determined by the IMF as given in Eq. (1). The IMF is a probability density function for the masses of formed stars, proportional to the relative numbers of stars formed with each mass, so the IMF needs to be weighted by mass in Eq. (8). This is achieved with a factor $\frac{m\Phi(m)}{\langle m\rangle_\Phi}$ is the distribution of the masses themselves, with the denominator being a normalising factor.

Figure (1) depicts how these formation and death rates enter into our compartment model. Eq. (8) results from combining the death rate of stars of mass m and the birth rate of stars of mass m. Eq. (9) comes from combining the total formation of stars from the ISM with the proportion p_{ret} of the total stellar mass that returns to the ISM. Eq. (10) comes from the proportion $(1 - p_{\text{ret}})$ of the total stellar mass that does not return to the ISM.

This model makes 4 major assumptions

1. Stars die at some rate, not according to their age.

2. The only factors of a galaxy are stars, the ISM, and stellar remnants.

3. The proportion of stars that return to the ISM is independent of mass.

4. Stars and stellar remnants do not accrete mass from the ISM.

Assumption 1 is examined further by D. C. Bruwel in their individual report. Assumption 2 is justified by classifying everything that isn't a star that can form stars as ISM, and the rest as remnants. Assumption 3 this is a limiting assumption, p_{ret} does depend on m as discussed by Hillebrandt and Leibundgut (2003), however this relationship between the mass of the star and the mass of the remnant is monotonic, making a straight line a subtile approximation for a wide range of stars, improvements could be made by incorporating the more complex mass dependence. Assumption 4 is a significant assumption, a more sophisticated model

could be made that includes accretion (i.e. absorb mass from interstellar medium), this is discussed later.

C. Mathematical Analysis

1. *Equilibrium and Stability Analysis*

To find the stable points of our model, we set the derivatives to zero. That is

$$f M_{\rm ISM}(t) \cdot \frac{m\Phi(m)}{\langle m \rangle_\Phi} - r(m)\,\rho(m,t) = 0 \tag{11}$$

$$-f M_{\rm ISM}(t) + p_{\rm ret} \int_{m_{\rm min}}^{m_{\rm max}} r(m)\rho(m,t)\,dm = 0 \tag{12}$$

$$(1-p_{\rm ret}) \int_{m_{\rm min}}^{m_{\rm max}} r(m)\rho(m,t)\,dm = 0 \tag{13}$$

Eq. (13) implies that

$$\int_{m_{\rm min}}^{m_{\rm max}} r(m)\rho(m,t)dm = 0 \implies \rho(m,t) = 0 \tag{14}$$

where the last implication comes from the fact that $\rho(m,t) \geq 0$ always. Combining this result with Eq. (12) implies that $M_{\rm ISM}(t) = 0$. As $M_{\rm R}$ is not involved in any of the differential equations, there is no restriction on its value. Simulations indicate that this point is a stable point. So, all the mass of the galaxy eventually becomes stellar remnants, as is expected to happen towards the end of the universe.

2. *Conservation of Mass*

Integrating Eq. (8) yields

$$\frac{dM_{\rm stars}(t)}{dt} = f M_{\rm ISM} - \int_{m_{\rm min}}^{m_{\rm max}} r(m)\rho(m,t)dm \tag{15}$$

Where $M_{\rm stars}$ is the total mass in stars given by $\int_{m_{\rm min}}^{m_{\rm max}} \rho(m,t)dm$, and we have used the fact that $\int_{m_{\rm min}}^{m_{\rm max}} m\Phi(m)dm = \langle m \rangle_\Phi$. Adding Eq. (15) to Eq. (9), and Eq. (10) yields.

$$\frac{dM_{\rm stars}(t)}{dt} + \frac{dM_{\rm ISM}(t)}{dt} + \frac{dM_{\rm R}(t)}{dt} = 0 \tag{16}$$

Which means that the total mass, $M_{\rm stars} + M_{\rm ISM} + M_{\rm R}$, is conserved as expected.

3. Initial Value Analysis

Starting with all the mass in M_{ISM}, and evaluating Eq. (8) yields

$$\frac{d\rho(m,t)}{dt} = fM_{\text{ISM}} \cdot \frac{m\Phi(m)}{\langle m \rangle_\Phi} \tag{17}$$

So at $t = \Delta t$ for some time just after $t = 0$ we have

$$\rho(m,t) = fM_{\text{ISM}} \cdot \frac{m\Phi(m)}{\langle m \rangle_\Phi} \Delta t \propto m\Phi(m) \tag{18}$$

As expected, the mass distribution for young galaxies is equal to the "initial mass function"

CHAPTER 3: RESULTS

D. Numerical Simulations

First, numerical simulations were run using Python without any data (data incorporated later). Various values of f and p_{ret} were used to see if the simulations agreed very roughly with the physics of the system. In the early universe, there were no stars (or star remnants), so all of the mass was in the interstellar medium. This was used for our initial condition, and then the differential equations were used to evolve the system for 20 billions of years, using relatively small discrete time steps in Python. 20 billion years was chosen as it is well known that the current age of the universe is 13.8 billion years (Planck Collaboration 2020), and thus it would be easy to compare to real data from today.

Fig. 2 shows one possible plot of stars, interstellar medium, and star remnants over 20 billion years. The parameters used were $p_{ret} = 0.85$ and $f = 1\,\text{Gyr}^{-1}$, corresponding to 85% of a dying star's mass returning to the interstellar medium, and all of the current interstellar mass going into star formation every billion of years. Star formation happens very rapidly at the beginning, as this is when most of the mass is in the interstellar medium. After 3 billion years, the rate at which stars form begins to slow down, until around 7 billion years where the mass in stars peaks, around 80% of the total mass of the system. After that, the mass in stars decreases extremely slowly. The mass in the interstellar medium follows an opposite trend. It first decays rapidly for the first 3 billion years, then continues to decrease slowly. The mass in the stellar remnants slowly builds up over time. This makes sense as mass enters

stellar remnants but does not leave it. Physically, this makes sense as we are currently in a period where there are many stars, little interstellar medium and a fair proportion of stellar remnants, and these are observations that can be made.

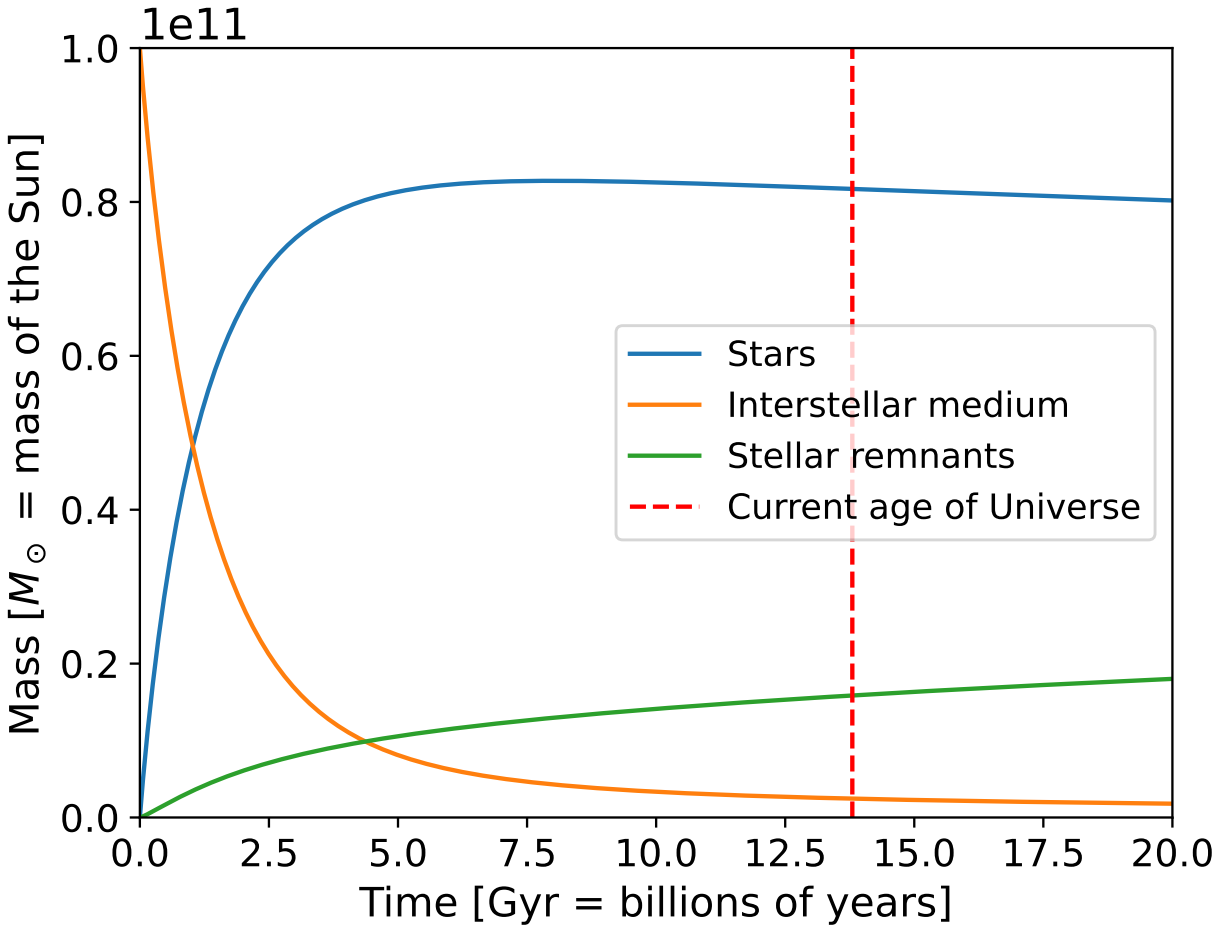


FIG. 2. Total mass in different forms over time for a Milky Way-like galaxy in a simulation using our model, with initial conditions $M_{\text{stars}} = 10^{11}\,M_\odot$, $M_{\text{ISM}} = M_{\text{R}} = 0$. $p_{\text{ret}} = 0.85$, $f = 1\,\text{Gyr}^{-1}$. The current age of the universe is marked at 13.8 Gyr (Planck Collaboration 2020).

Fig. 3 shows a log-log plot of the mass function over time for the same parameters above. The mass functions have discontinuous derivatives at two point, but this is because the IMF has the form of piecewise power-law, as seen earlier. Over time, the higher mass stars start to decrease. This is consistent with physics, as higher mass stars die more quickly because they go through their fuel more quickly to remain in hydrostatic equilibrium. However, because of star formation, the higher mass stars do not completely die off. This is a significant change

to the simple stellar population explored in Verro *et al.* (2022). Correspondingly, there is a slight increase in lower mass stars over time. Visibly, this is not so noticeable on a log-log plot, however this must be the case, as the mass functions are probability density functions, and the high mass stars decrease.

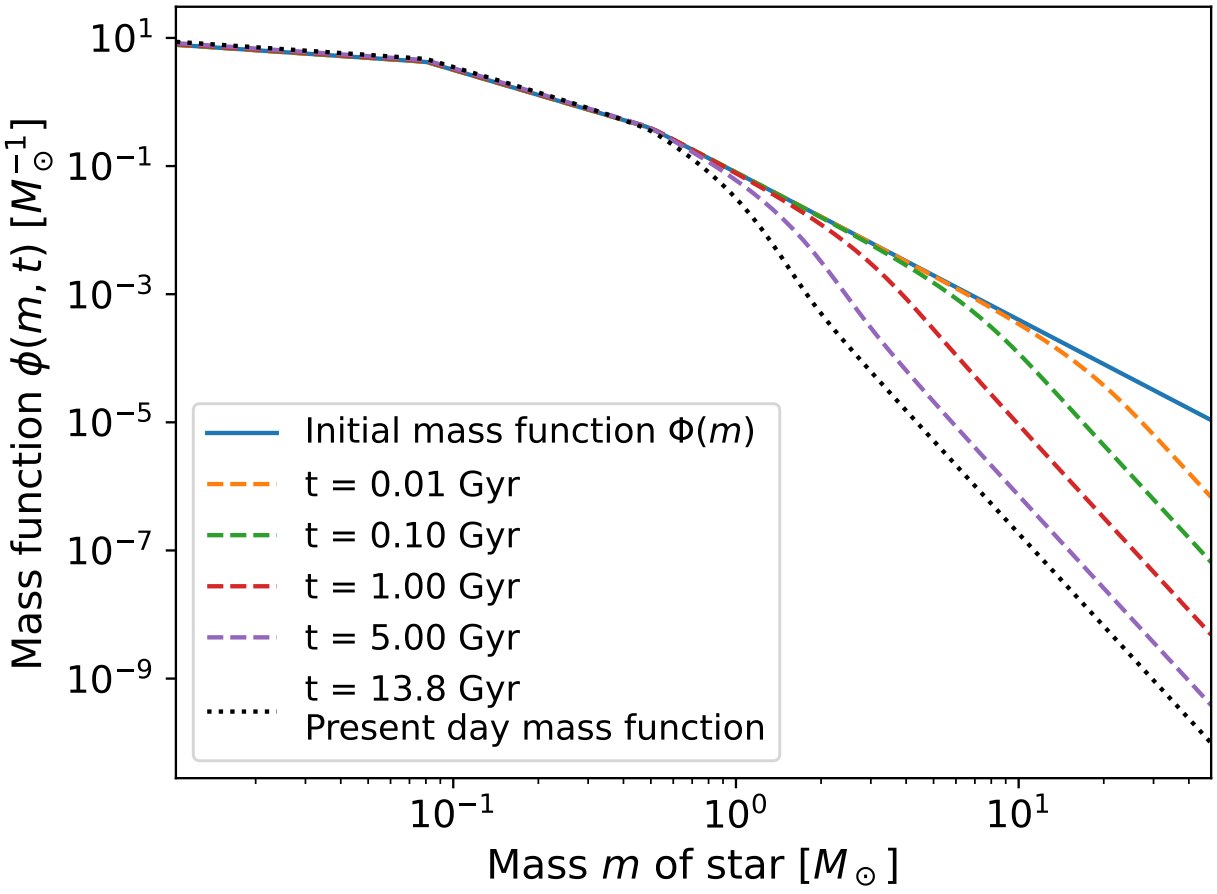


FIG. 3. Time-evolved mass functions for the simulation described in Figure 2, showing also the Kroupa (2001, 2002) IMF given in Eq. (1).

Fig. 4 shows a log-log plot of mass functions of the present day for various different values of f. Even though f changes over seven orders of magnitude, the mass functions do not vary much. All of them are consistent with higher mass stars dying off much faster than lower mass stars.

Fig. 5 shows a log-log plot of mass functions of the present day for various different values of p_{ret}. This time, the mass function does change considerably for lower values of p_{ret}. This shows that when there is little mass returning back to the ISM, higher mass stars do not get

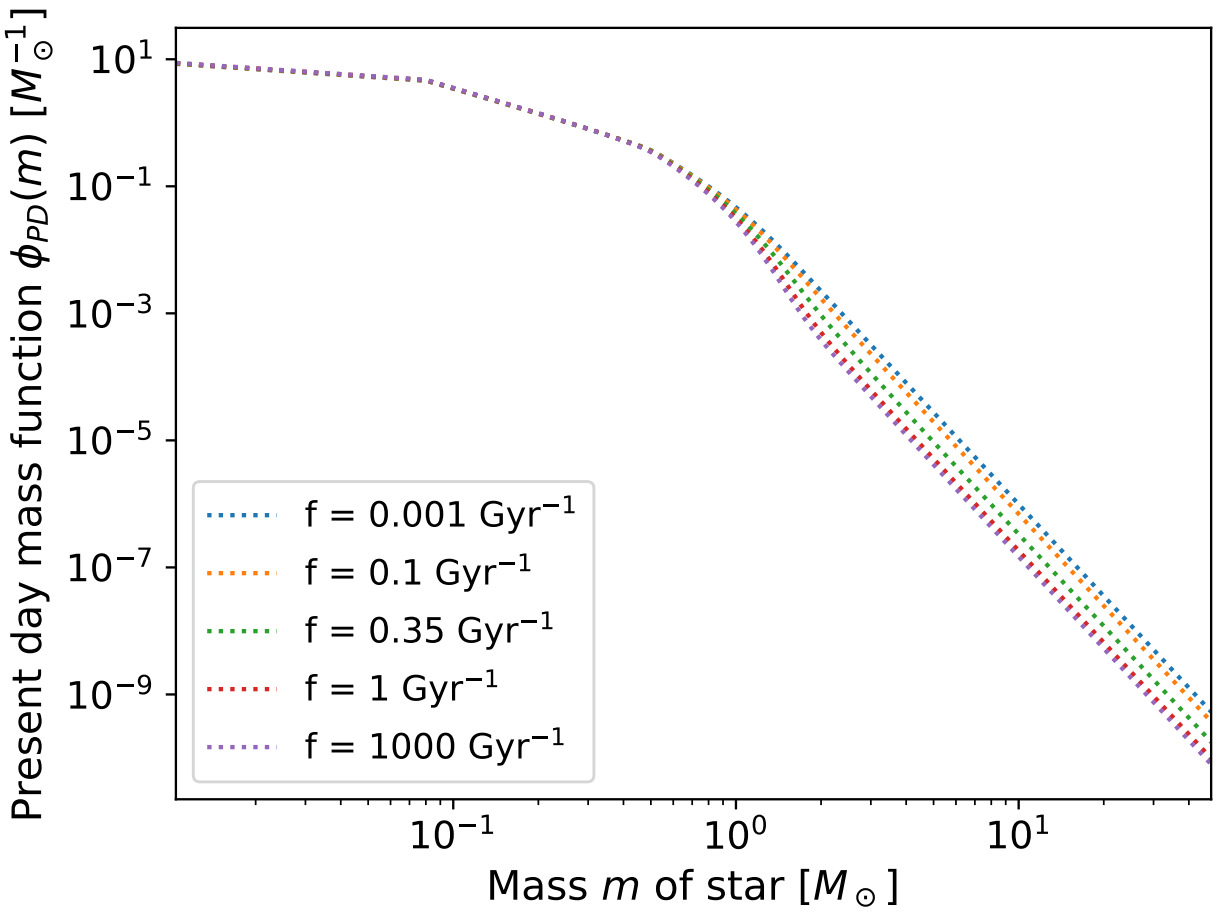


FIG. 4. Present day mass function predictions from model with different stellar formation rates f (fixed $p_{\text{ret}} = 0.85$).

reformed as fast as they die, which is different to the effect that the formation rate had on the present day mass function.

E. Data Collection

After numerical simulations, data was collected and fitted to find the best parameters for f and p_{ret}.

Data was collected from the sixteenth data release of the Sloan Digital Sky Surveys Ahumada *et al.* (2020) which is an online data catalogue of various celestial objects such as galaxies, and their properties. This data was then put into Python and cleaned to be used for fitting.

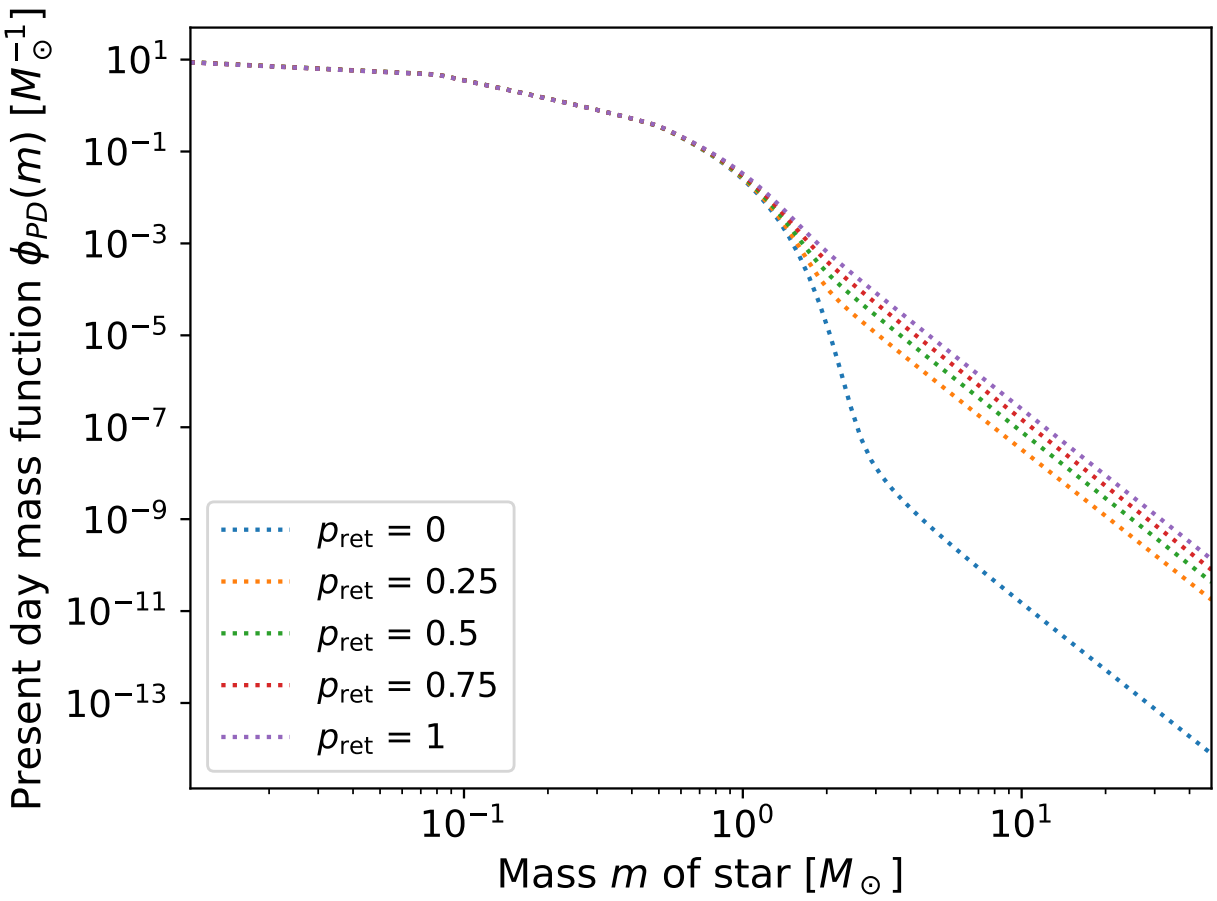


FIG. 5. Present day mass function predictions from model with different proportions p_{ret} returned to ISM upon death of star (fixed $f = 1\,\text{Gyr}^{-1}$).

Notably, we used it to find various galaxies and get their spectra, which is the distribution of emitted light from that galaxy. This spectra will be essential for fitting, which is described later. Due to the expansion of the universe, all objects are moving away from each other, and so the wavelengths of the emitted light gets stretched and turns more red. This is known as redshift. In fact, this redshift can be used to determine how long ago the light was emitted (lookback time), as galaxies further away would take longer for the light to reach Earth. This was calculated in Python, using the astropy package. This then means it is possible to figure out how old the galaxy is, as all galaxies were born at the same time shortly after the Big Bang. The spectra was also adjusted to account for this redshift, to get the true distribution of emitted light (as opposed to the light observed on Earth). Sixteen different galaxies of various redshifts (from 0.05 to 0.7) were used, and put into a table with their

redshifts, lookback time, age of galaxy when light was emitted, spectra, and de-redshifted spectra.

F. Fitting

Equipped with this data, we now needed a way to compress our stellar mass distribution into a galactic spectrum, which could be compared to the data obtained in Section E. We first detail the method of this compression, then explain how fitting the model parameters was completed.

1. Compression

The idea here is to think of Figure 3 as detailing distribution of stars in a galaxy, each of which will emit its own spectrum, and then find the weighted sum of these spectra.

A star can be approximated as a 'blackbody' to a high degree of precisionSeeds and Backman (2016). A blackbody is an idealised object in physics with a known analytic formula (Carroll and Ostlie 2007):

$$I(\lambda) = 2hc^2 \frac{\lambda^{-5}}{e^{hc/\lambda kT} - 1} \tag{19}$$

Where I is the intensity of light, $h = 6.626 \times 10^{-34} Js$ is Planck's constant, $c = 3 \times 10^8 ms^{-1}$ is the speed of light, $k = 1.381 \times 10^{-23} JK^{-1}$ is Boltzmann's constant, and T is the temperature of the emitting object in Kelvin - here, the surface of the star.

To use this we need the temperature T of the star, which can be found from the Stefan-Boltzmann law relating radius R and luminosity L of a star to its temperature (Carroll and Ostlie 2007)

$$L = 4\pi\sigma R^2 T^4 \tag{20}$$

where $\sigma = 5.670 \times 10^{-8}\,\mathrm{W\,m^{-2}\,K^{-4}}$ is the Stefan-Boltzmann constant, the mass-luminosity relation (Eq. (4)), and the mass-radius relation (5).

The latter two of these are empirical fits that only hold for MS stars as discussed in Section B 2. Combining these gives (for m the mass of a star, $T_\odot$ and $M_\odot$ the solar temperature

and mass respectively, and T)

$$\frac{T}{T_\odot} = \left(\frac{m}{M_\odot}\right)^{0.475} \tag{21}$$

From temperature, we can find the blackbody spectrum (Eq. (19)). In practice this is computed on a discrete sample of wavelengths λ - exactly those found in a given galaxy spectrum - and then scaled by an arbitrary constant A which depends on the galaxy, and incorporates: unit conversions of intensity, the number of stars in the galaxy, the attenuation of light by dust and by distance (according to the inverse square law Seeds and Backman (2016)) on its path to Earth. This is set as a vector of parameters $\vec{A}$ is optimized as a nested optimization problem within each calculation of the loss function.

2. Optimisation

Our task is then simply to use this modeled spectrum by comparing it to fit the parameters, by comparing it to the data of the galactic spectrum obtained earlier. To do this we develop a simple squared-difference loss function, modeled discretely by

$$\sum_i \sum_\lambda \left| I^{p_{\text{ret}},f,A_i}_{\text{model}}(\lambda) - I^{p_{\text{ret}},f,A_i}_{\text{data}}(\lambda) \right|^2$$

where the outer sum is over all galaxies, indexed by i, and the inner sum over wavelength points from the dataset for each particular galaxy. We then set this to be minimized for $p_{\text{ret}} \in (0,1)$, $f \geq 0$, with the 'trust-constr' from Virtanen *et al.* (2020).

In practice, the runtime of the code is very poor leading to constraints on the optimization technique used. The final optimized parameters are unlikely to be global optima.

After doing this, we obtain the parameters

$$p_{\text{ret}} = 13.0\% \tag{22}$$

$$f = 41.9\ . \tag{23}$$

p_{ret} is slightly lower than expected, as when a high mass star dies - for instance with mass $25M_\odot$ - it becomes a neutron star with mass on the order 1 to 3 $M_\odot$ Carroll and Ostlie (2007), and thus the other $\approx 85\%$ has become interstellar remnant - i.e. $p_{\text{ret}} \approx 0.85$. Our model implies a lower return rate of stellar matter to the ISM - but on the same order of magnitude to 0.85, which is at least reassuring. Exact values are not widely accepted and

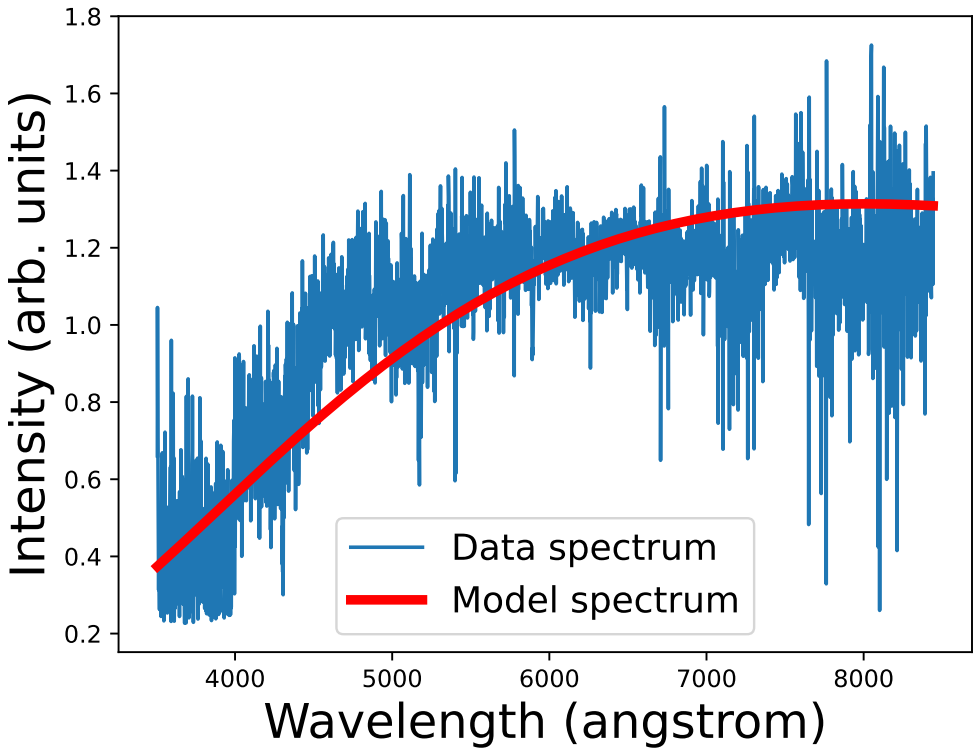


FIG. 6. Example fit for galaxy from catalogue, with parameters from 22, galaxy is "SDSS J025831.54+005626.5"

difficult to pin down due to the intricately complex processes occurring here (Seeds and Backman 2016). The same is true for f, which is somewhat higher, but on the same order of magnitude, compared to approximate expectations (Seeds and Backman 2016).

An example plot of the model's result for a galactic spectrum with these parameters is seen in Figure 6. Here we can see that the fit is quite reasonable, adhering to the broad shape of the galactic spectrum. If the model spectrum increased more steeply at lower wavelengths and more slowly at higher wavelengths, the fit would be closer.

G. Assumptions/limitations

Since the system we are trying to model is extremely complicated, there are many assumptions and simplifications that had to be made to make the model feasible. Of course, we have tried to limit them, or make reasonable assumptions that would be reflected in the real system. Here we discuss the most significant assumptions, and the justifications behind

them.

In the results above, the mass is converted to spectra, through mass, temperature and luminosity relations. However, these relations assumes that all stars are main sequence stars. This is a reasonable assumption, as roughly 90% of stars are in the main sequence (Seeds and Backman 2016). However, when a star is not in the main sequence (mostly red giants), then these relations are no longer very accurate. It turns out that the luminosity of these stars are significantly higher, and the temperature is slightly lower. Specifically, the luminosity being higher may have a noticeable impact on the emitted spectra of a galaxy, especially for wavelengths that correspond to red. However, determining temperature and luminosity from mass for non-main sequence stars is complicated and computationally intensive. As such, we could not avoid this, however because only 10% of stars are not in the main sequence, this assumption still holds relatively well.

Another assumption we have made is that galaxies do not interact with other galaxies. In reality, galaxies may merge together, combining their stellar populations and interstellar media (Seeds and Backman 2016). In fact, elliptical galaxies are made from galaxy merges, which turns out to follow the simple stellar population from Verro *et al.* (2022) relatively well. Furthermore, galaxies that merge can affect the ongoing star formation rate in the combined galaxy, although Pearson *et al.* (2019) find this effect to be small. Similarly, our model does not account for interactions between stellar populations. For example, stars will not form around existing stars, as the gravitational attraction from the existing star will be much stronger than the gravity between interstellar medium required to form a new star. In some cases, it is also possible for two stars to merge together. Although our model does not incorporate these, we believe that these situations are minor relative to the size of the galaxies, and so our assumptions are still reasonable.

One simplification mentioned in the model was that stars and stellar remnants do not accrete mass (i.e. absorb mass from interstellar medium). This is not the case especially for stellar remnants. Furthermore, the more mass accreted, the heavier the star/stellar remnant gets, which creates stronger gravity which leads to more accretion of mass. Similarly, intergalactic medium could also be incorporated into the model, which is medium between galaxies. This medium could be absorbed by galaxies, which becomes interstellar medium for that galaxy. We believe that this is the change which may potentially have a significant

impact on the results. Both of these affects could be incorporated into the model by adding more terms in the system of differential equations.

CONCLUSIONS

The mathematical analysis and simulations indicate that our model produces results that have dynamics consistent with the current accepted understanding of the universe. Its agreement with expectations from physics, both from qualitative inspection of mass distributions obtained in section C 3, such as the tapering off of mass distribution for the high-mass stars with their short lifetime. The results from fitting our model to data are limited, while the initial fitting to data shows promise, a larger data set and more sophisticated optimization techniques would be required to confirm these results and 'hone in on' the exact parameters.

A number of limitations of our model have been outlined throughout the report, addressing these limitations would be the most significant improvement that could be made to the model. More pragmatically, evolving the model for some initial set of parameters is incredibly time consuming (taking around 6 seconds on an 11th generation 2.8GHz Intel i7 core). Improving the efficiency of this would allow for more complex models, larger data sets, and the possibility of using global optimization techniques.

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